Case Study 01: ATM System Design

Cover Page

Title: Case Study 01 – ATM System Design
Domain: Software Engineering / System Design
Focus: Secure, Scalable Automated Teller Machine (ATM) System

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1. Executive Summary

The Automated Teller Machine (ATM) system is a critical banking application that allows customers to perform essential financial transactions without visiting a bank branch. This case study presents a complete design of a modern ATM system, covering functional behavior, system architecture, security controls, and testing strategies.

The proposed system supports transactions such as cash withdrawal, balance inquiry, cash deposit, fund transfer, and PIN change. It also enables bank administrators to perform maintenance and cash replenishment activities. The design follows a layered MVC (Model–View–Controller) architecture to ensure scalability, maintainability, and reliability.

The system is designed to operate 24/7 with a target availability of 99.9%, comply with PCI DSS standards, and protect customer data using strong encryption and fraud-prevention mechanisms.

Visual – System Overview Diagram (Conceptual):
Customer → ATM Machine → Secure Network → Bank Server → Database

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2. Requirements

2.1 Functional Requirements

- Authenticate users using ATM card and PIN.
- Allow customers to withdraw cash.
- Allow customers to deposit cash.
- Display account balance.
- Transfer funds between accounts.
- Change ATM PIN.
- Print transaction receipts.
- Retain card after three invalid PIN attempts.
- Allow bank admin to perform maintenance and cash loading.

2.2 Non-Functional Requirements

- Transaction response time less than 30 seconds.
- Handle 500+ transactions per day per ATM.
- Operate continuously (24/7 availability).
- Support multiple languages.
- Provide accessibility for disabled users.
- Ensure 99.9% system uptime.

2.3 Requirement Classification Table

Requirement Type	Examples	Priority
Functional	Withdraw Cash, Deposit Cash	High
Non-Functional	<30s response time	High
Security	PIN Encryption	Critical

Visual – Requirement Classification Diagram:
Box diagram separating Functional, Non-Functional, and Security requirements.

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3. Actors & Use Cases

3.1 Actors

- **Customer:** Uses ATM for banking transactions.
- **Bank Admin:** Performs maintenance and cash management.
- **Bank Server:** Validates transactions and authenticates users.

3.2 Use Cases

- Authenticate User
- Withdraw Cash
- Deposit Cash
- Check Balance
- Transfer Funds
- Change PIN
- System Maintenance

3.3 Use Case Diagram (Textual Representation)

-**Visual – Use Case Diagram:**
Stick-figure actors connected to oval-shaped use cases with include/extend relationships.

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4. Architecture

4.1 Architectural Style

The ATM system follows a **Layered MVC Architecture**:

- **Model Layer:** Handles data and business entities.
- **View Layer:** User interface components.
- **Controller Layer:** Transaction coordination and logic.

4.2 Architecture Components

Layer	Components	Responsibility
Presentation	Screen, Printer	User Interaction
Business	Transaction Manager	Validation & Logic
Data	Account DB	Data Storage

Layer	Components	Responsibility
External	Bank Server	Authentication

4.3 Architecture Diagram (Textual)

Visual – Architecture Diagram:
Layered boxes with arrows showing data flow to external bank server.

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5. Design Flow

5.1 Transaction Flow Description

1. Customer inserts ATM card.
2. System prompts for PIN.
3. PIN is validated with bank server.
4. Customer selects transaction type.
5. Transaction is processed.
6. Cash is dispensed (if applicable).
7. Receipt is printed.
8. Card is ejected and session ends.

5.2 Error Handling

- Invalid PIN: Retry allowed up to 3 times.
- Insufficient funds: Transaction cancelled.
- Cash unavailable: Admin notified.

5.3 Flow Chart (ASCII Flow Diagram)

Visual – Flow Chart:
Standard flowchart with decision diamonds and process blocks.

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6. Security

6.1 Security Measures

- PIN encryption using Triple DES.
- Secure communication using TLS 1.3.

- EMV chip-based card authentication.
- Role-based access control.
- Tamper-evident ATM hardware.

6.2 Threat Mitigation

- Brute-force prevention using attempt limits.
- Real-time fraud monitoring.
- Automatic card retention on suspicious activity.

6.3 Security Diagram (Textual)

[ATM] --Encrypted--> [Firewall] --TLS--> [Bank Server]

Visual – Security Diagram:
Lock and shield icons on communication paths.

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7. Testing

7.1 Testing Strategy

- Unit Testing
- Integration Testing
- System Testing
- User Acceptance Testing (UAT)
- Security Testing
- Load Testing

7.2 Testing Pyramid Diagram (Textual)

UAT

System Test

Integration Test

Unit Test (Base)

7.3 Testing Table

Test Type	Tools	Goal
Unit	JUnit	90% Coverage

Test Type	Tools	Goal
Security	OWASP ZAP	Vulnerability-Free
Load	JMeter	<30s Response

Visual – Testing Diagram:
Testing pyramid illustrating coverage levels.

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8. Conclusion

The ATM System Design presented in this case study provides a secure, reliable, and scalable solution for modern banking needs. By adopting a layered architecture, strong security mechanisms, and comprehensive testing strategies, the system ensures high availability and customer trust.

Future enhancements may include biometric authentication, contactless transactions, and AI-based fraud detection. The proposed design is suitable for real-world deployment and aligns with industry standards.
